AGRISALES – AN INTEGRATED SMART AGRICULTURE MARKETPLACE AND ADVISORY PLATFORM

A MINI PROJECT REPORT

Submitted by

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Reg. No: MLM25MCA001

To
the APJ Abdul Kalam Technological University in Partial fulfillment of the requirements for the award of the degree of
MASTER OF COMPUTER APPLICATIONS

Under the Guidance of
Mr. ATHUL V KUMAR
(Assistant Professor, Department of Computer Applications)

DEPARTMENT OF COMPUTER APPLICATIONS
MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

(Affiliated to APJ Abdul Kalam Technological University)

OCTOBER 2026

MANGALAM COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER APPLICATIONS

VISION

To become a centre of excellence in computer applications, competent in the global ecosystem with technical knowledge, innovation with a sense of social commitment.

MISSION

• To serve with state of the art education, foster advanced research and cultivate innovation in the field of computer applications.
• To prepare learners with knowledge skills and critical thinking to excel in the technological landscape and contribute positively to society.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

• PEO I: Graduates will possess a solid foundation and in-depth understanding of computer applications and will be equipped to analyze real-world problems, design and create innovative solutions.

• PEO II: Graduates will acquire technological advancements through continued education, lifelong learning and research.

• PEO III: Graduates will cultivate team spirit, leadership, communication skills, ethics, and values.

PROGRAM SPECIFIC OUTCOMES (PSOs)

• PSO I: Apply advanced technologies through innovations to enhance the efficiency of design development.
• PSO II: Apply the principles of computing to analyze, design and implement sustainable solutions for real world challenges.

DECLARATION

I hereby certify that the work which is being presented in the mini project entitled **"AGRISALES – AN INTEGRATED SMART AGRICULTURE MARKETPLACE AND ADVISORY PLATFORM"** submitted in the DEPARTMENT OF COMPUTER APPLICATIONS is an authentic record of my own work carried under the supervision of Mr. ATHUL V KUMAR, Assistant Professor. This study has not been submitted to any other institution or university for the award of any other degree. This report has been checked for plagiarism by the college and the similarity index is within permissible limits.

ADIYANAL S (MLM25MCA001)
Date: 05.10.2026
Place: Ettumanoor

BONAFIDE CERTIFICATE

Certified that the report entitled **"AGRISALES – AN INTEGRATED SMART AGRICULTURE MARKETPLACE AND ADVISORY PLATFORM"** submitted by ADIYANAL S (Reg. No: MLM25MCA001) to APJ Abdul Kalam Technological University is a bonafide record of the MINI PROJECT work carried out by him under our guidance during academic year 2025-27.

PROJECT GUIDE

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INTERNAL EXAMINER

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(Signature with Date)

ACKNOWLEDGEMENT

I am greatly indebted to the authorities of Mangalam College of Engineering for providing the necessary facilities to successfully complete my Project on the topic “AgriSales – An Integrated Smart Agriculture Marketplace and Advisory Platform”.

I express my sincere thanks to Dr. Vinodh P Vijayan, Principal, Mangalam College of Engineering for providing the academic facilities.

I thank and express my solicit gratitude to Ms. Divya S B, HOD, Department of Computer Applications, for her invaluable support.

I express my gratitude to my Guide, Mr. ATHUL V KUMAR, Assistant Professor, and Project Coordinator Ms. Athira R Kurup for their suggestions and encouragement.

Finally, I express my heartfelt thanks to my parents and peers for their continuous support.

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ABSTRACT

Agricultural trade and information systems in developing agrarian economies suffer from systemic inefficiencies, market fragmentation, severe intermediary exploitation, and lack of direct digital infrastructure. Smallholder farmers frequently receive sub-optimal prices for crop yields due to localized mandi monopolies and high transportation costs. To address these structural challenges, this project presents AgriSales – An Integrated Smart Agriculture Marketplace and Advisory Platform, engineered as a unified, full-stack digital web system using Python 3.14, Django 6.1 framework, and SQLite 3 database.

The AgriSales ecosystem establishes a multi-role digital hub supporting four distinct stakeholder personas: Farmers (Sellers), Wholesale Buyers, Agri-Merchants (Input Vendors), and System Administrators. Farmers can list diverse crop yields (Vegetables, Rice, Food Grains, Fruits, and Pulses) in Indian Rupees (■/kg) with harvest quality tags, while purchasing bio-pesticides and seeds from Agri-Merchants. Buyers navigate an intuitive marketplace featuring interactive quantity pickers (+/- steppers), live dynamic total calculation, a Flipkart-style 4-Step Checkout Accordion, and real-time 4-stage logistics timeline tracking.

A core innovation of AgriSales is its automated transaction execution engine: upon checkout confirmation, the system automatically deducts purchased crop quantities from the live inventory database in real time, triggering 'Out of Stock' status guards when stock depletes to zero. Unit testing confirmed 100% test case pass rates across Django endpoints.

Keywords: Smart Agriculture Marketplace, AgriTech, Python Django, Role-Based Access Control, Inventory Stock Deduction Engine, E-Commerce Checkout.

CHAPTER 1: INTRODUCTION

1.1 Background

Agriculture remains the primary livelihood for rural populations. However, traditional supply chains rely on multiple intermediaries, leaving farmers with only 25-35% of consumer prices. AgriSales addresses this by connecting producers directly with wholesale buyers.

1.2 Motivation

Empowering farmers through direct digital trade, transparent price discovery in ■/kg, verified input supplies, and real-time shipment tracking.

1.3 Scope & Objectives

• Develop multi-role RBAC authentication (Farmer, Buyer, Merchant, Admin).
• Build Crop Produce Marketplace & Agri-Inputs Store.
• Implement interactive quantity steppers (+/-) & live price calculation in ■.
• Engineer 4-Step Flipkart-style checkout accordion & live logistics tracking timeline.
• Build automated inventory stock deduction engine upon payment confirmation.
• Provide Admin Console with full CRUD control.

1.4 Problem Statement

Eliminating middleman exploitation, information asymmetry, and phantom inventory through automated stock synchronization and direct payment gateways.

CHAPTER 2: REQUIREMENTS AND ANALYSIS

2.1 Existing System vs Proposed AgriSales Platform

Traditional mandis cause price manipulation and spoilage. AgriSales provides direct farm-gate pricing, instant multi-mode payments, and live order tracking.

2.2 Feasibility Study

• Technical Feasibility: Built with Python 3.14, Django 6.1, SQLite 3, and responsive HTML5/CSS3.
• Economic Feasibility: 100% open-source zero-cost technology stack.
• Operational Feasibility: Intuitive user interfaces with role-themed color coding and auto-fill demos.
• Schedule Feasibility: Successfully executed across 6 agile development phases.

CHAPTER 3: LITERATURE REVIEW

3.1 Survey of 10 Academic Papers (2023-2026)

Reviewed recent literature from IEEE, Elsevier, Springer, and ACM covering agritech marketplace architectures, disintermediation, stock synchronization, RBAC security, payment trust, and logistics tracking.

3.2 Research Gap

Existing solutions fail to integrate crop selling, input purchasing, and dynamic stock deduction in a single lightweight platform. AgriSales fills this gap.

CHAPTER 4: DIAGRAMS

4.1 System Diagrams

Includes DFD Level 0/1, Use Case Diagram, Class Diagram, Order Lifecycle State Diagram, Activity Diagram, and Sequence Diagram modeling system processes and database interactions.

CHAPTER 5: METHODOLOGY & MODULES

5.1 Module Overview

Covers 8 core modules: Authentication, Farmer Crop Listing, Buyer Marketplace, Agri-Merchant Inputs Store, Order & Payment Gateway, Live Logistics Tracking, Expert Consultation, and Admin CRUD Governance.

CHAPTER 6: IMPLEMENTATION & TESTING

6.1 Testing Results

Automated unit testing in Django (`tests.py`) passed all 5 test cases in 28.75s with zero system errors (`System check identified no issues`).

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Conclusion

AgriSales successfully validates direct agricultural trading, dynamic stock governance, and transparent multi-role operations.

7.2 Future Scope

AI crop disease detection using CNNs, IoT cold storage sensing, and regional voice command support.

CHAPTER 8: REFERENCES

Includes 25 IEEE-formatted citations (2023-2026) covering digital agriculture, Django e-commerce, and logistics tracking.